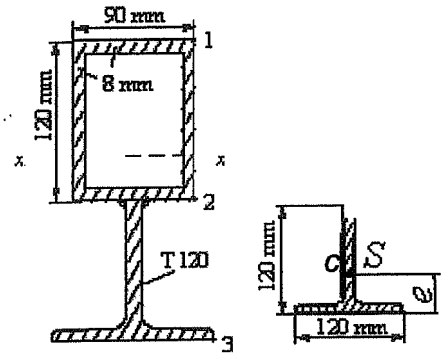
Copy of Tutorial Solutions in Resource Centre for

Tutorial 1: Centroids

Determine the y-co-ordinate of the centroid from base for the following three profiles by using the table technique. *(Assume missing units in mm)*

(S1.1)



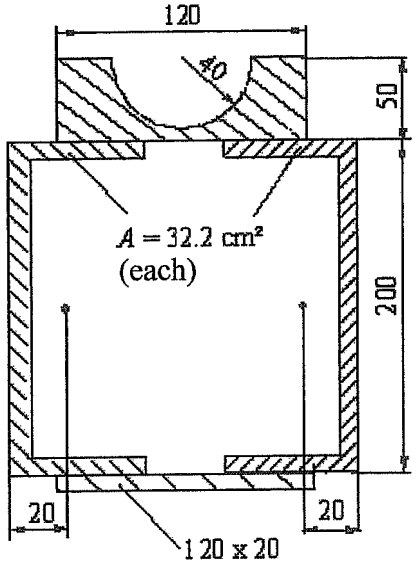
$T120$

$A = 29.6\ \text{cm}^2$

$e = 3.28\ \text{cm}$

(Ans. 10.8 cm)

(S1.2)



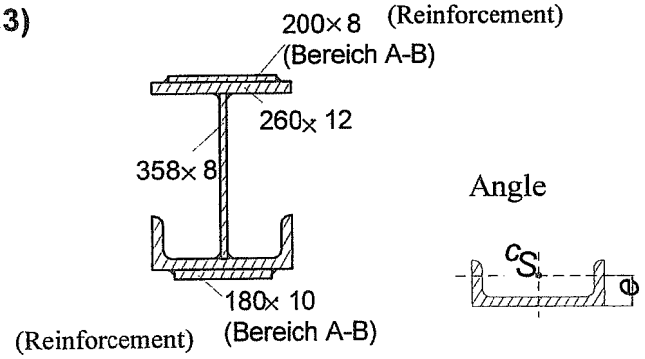
for a semi-circle :

$y_c = \dfrac{4 \cdot r}{3 \cdot \pi}$

$t = 1\text{cm}$
(thickness of the two angles)

(Ans. 13.2 cm)

(S1.3)



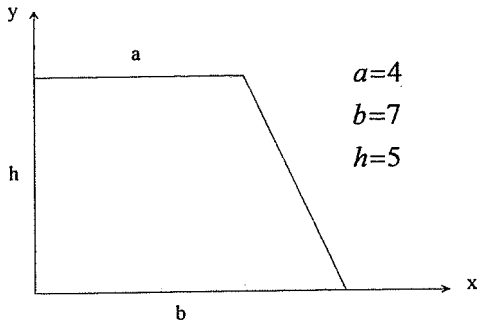
$A = 48.3\ \text{cm}^2$

$e = 2.36\ \text{cm}$

$t = 1\text{cm}$
(thickness of angle)

(calculation to be performed with and without reinforcements) *(Ans. 16.8 and 18.1 cm)*

(S1.4)



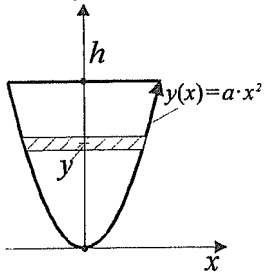
$a=4$
$b=7$
$h=5$

All dimensions are in cm.

Calculate Y_c and X_c
(Ans. $X_c = 2.82$ and $Y_c = 2.27$ cm)

(S1.5)

Find the location of centriod (y_c) of a quadratic parabola (Figure 3.5) of height "h" using direct integration method. Assume $y(x) = ax^2$, where "a" is a constant.
(Hint. First-find the area of parabola by direct integration)



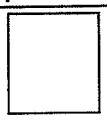
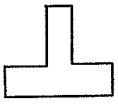
(Figure 3.5)

(Ans. 3h/5)

Solutions for Tutorial 1: Centroids

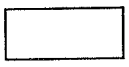
The calculations were performed in units of cm^n.

(S1.1)

profile:	y	A	$y \cdot A$
	18	108	1944
	18	−76.96	−1385
	3.28	29.6	97
$\sum$		60.64	656

$$\underline{\underline{y_c}} = \frac{\sum y_i A_i}{\sum A_i} = \frac{656\ \mathrm{cm}^3}{60.64\ \mathrm{cm}^2} = \underline{\underline{10.82\ \mathrm{cm}}}$$

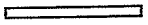
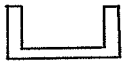
(S1.2)

profile:	y	A	$y \cdot A$
	25.3	−25.1	−635
	24.5	60	1470
	12	64.4	772.8
	1	24	24
$\sum$		123.3	1632

$$\underline{\underline{y_c}} = \frac{\sum y_i A_i}{\sum A_i} = \frac{1632\ \mathrm{cm}^3}{123.3\ \mathrm{cm}^2} = \underline{\underline{13.2\ \mathrm{cm}}}$$

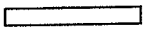
(S1.3)

without reinforcement:

profile:	y	A	$y \cdot A$
	37.4	31.2	1167
	18.9	28.64	541
	2.36	48.3	114
$\sum$		108.14	1822

$$\underline{\underline{y_c}} = \frac{\sum y_i A_i}{\sum A_i} = \frac{1822\ \text{cm}^3}{108.14\ \text{cm}^2} = \underline{\underline{16.84\ \text{cm}}}$$

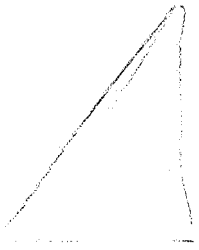
with reinforcement:

profile:	y	A	$y \cdot A$
	39.4	16	630.4
	38.4	31.2	1198
	19.9	28.64	570
	3.36	48.3	162.3
	0.5	18	9
$\sum$		142.14	2569.7

$$\underline{\underline{y_c}} = \frac{\sum y_i A_i}{\sum A_i} = \frac{2569.7\ \text{cm}^3}{142.14\ \text{cm}^2} = \underline{\underline{18.1\ \text{cm}}}$$

(S1.4)

profile:	x	y	A	$x \cdot A$	$y \cdot A$
	2	2.5	20	40	50
	5	1.67	7.5	37.5	12.5
$\sum$			27.5	77.5	62.5



$$\underline{\underline{x_c}} = \frac{\sum y_i A_i}{\sum A_i} = \frac{77.5\ \text{cm}^3}{27.5\ \text{cm}^2} = \underline{\underline{2.82\ \text{cm}}}$$

$$\underline{\underline{y_c}} = \frac{\sum y_i A_i}{\sum A_i} = \frac{62.5\ \text{cm}^3}{27.5\ \text{cm}^2} = \underline{\underline{2.27\ \text{cm}}}$$

(S1.5)

We can evaluate the area of quadratic parabola by summing up all contributions from the slices shown in Figure 3.5. In other words, we integrate as follows:

$$\text{Area of parabola} = A = 2\int_{y=0}^{y=h} x\,\mathrm{d}y = \frac{4}{3}\frac{h^{3/2}}{\sqrt{a}}$$

Also:

$$\mathrm{d}A = 2x\,\mathrm{d}y = \frac{2y^{1/2}\,\mathrm{d}y}{\sqrt{a}}$$

$$\Rightarrow \quad \int_A y\,\mathrm{d}A = \frac{2}{\sqrt{a}}\int_{y=0}^{h} y^{3/2}\,\mathrm{d}y = \frac{4h^{5/2}}{5\sqrt{a}}$$

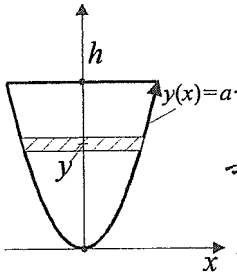


Fig. 3.5: Centroid of a parabola.

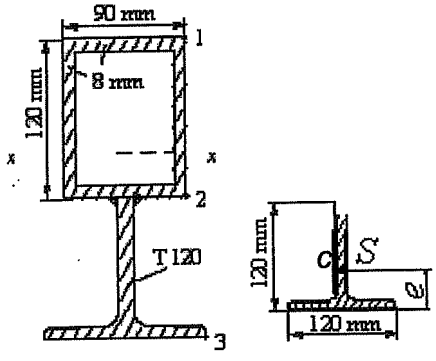
This results in:

$$y_c = \frac{1}{A}\int_A y\,\mathrm{d}A = \frac{3}{5}h$$

Tutorial 2: Second Moment of Area

Determine the second moment of area and section Moduli at the top and bottom (for **S2.1** to **S2.3**) of the following profiles (Use values of Y_c from Tutorial 1).

(S2.1)



$T120$

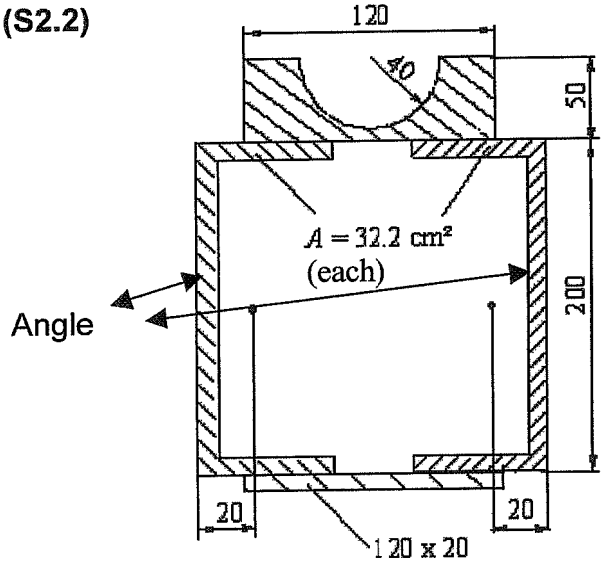
$A = 29.6\ cm^2$

$e = 3.28\ cm$

I_o = 366cm^4

(Ans. I_o = 4250 cm^4, Z =322 and 393cm^3)

(S2.2)



for a semi-circle :

$y_c = \frac{4 \cdot r}{3 \cdot \pi}$

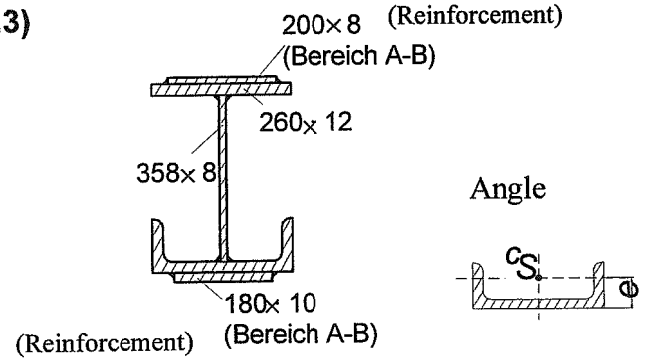
t = 1cm (thickness of the two angles)

neglect I_o for semi-circle

I_o per angle = 466.5cm^4

(Ans. I_o = 8709 cm^4 Z =631 and 660 cm^3)

(S2.3)



For Angle

I_o = 317cm^4

A = 48.3 cm^2

e = 2.36 cm

t = 1cm (thickness of angle)

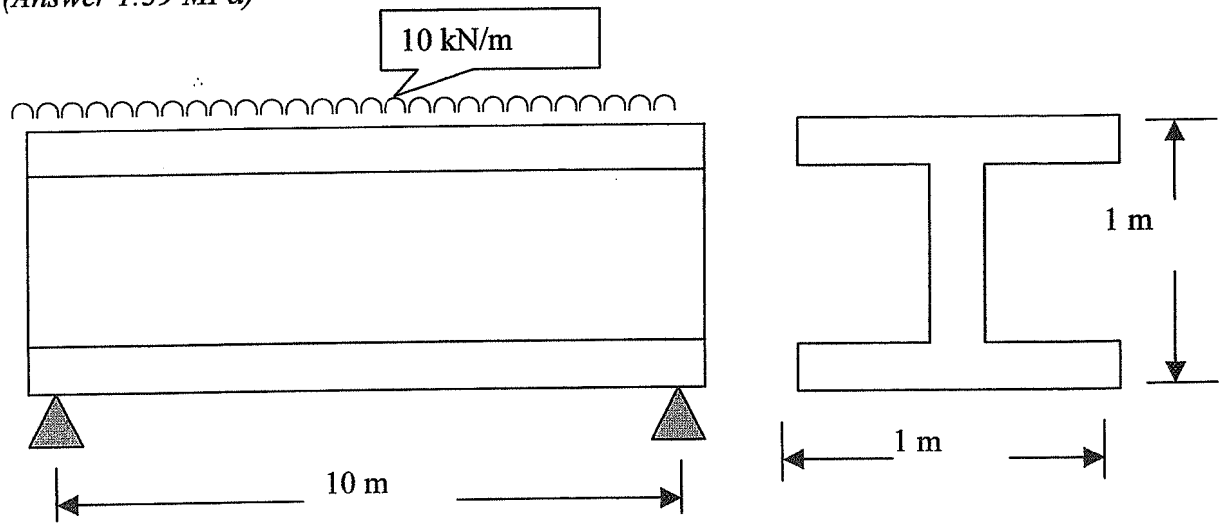
(calculation to be performed with and without reinforcements)

(Ans. I_o = 26815 cm^4, Z =1591 and 1268 cm^3 without Reinforcements)

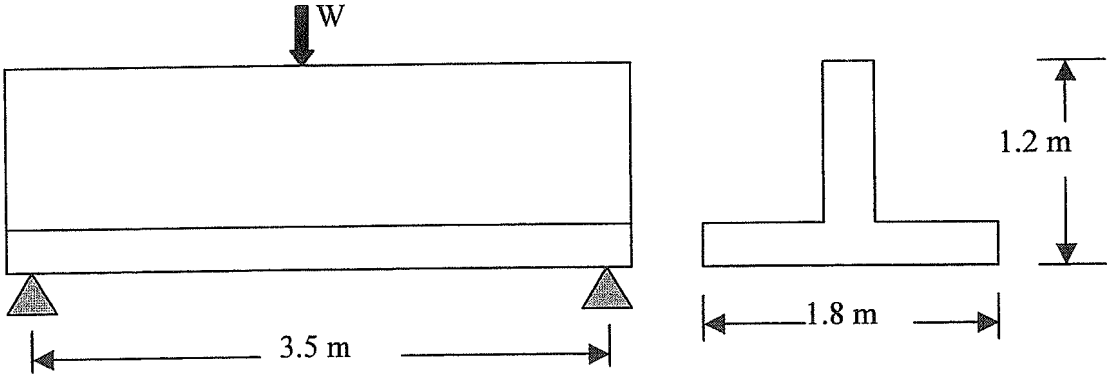
(Ans. I_o = 39655 cm^4, Z =1827 and 2191 cm^3 with Reinforcements)

(S2.4)

A simply supported beam of I section (shown in Figure below) is simply supported at both ends with a clear span (L) of 10m. A (UDL) uniformly distributed load (ω) of 10 kN/m acts on its span. If the thickness of web and each flange is 0.1m, calculate the maximum stress produced in beam material due to bending.
(Hint-Maximum bending moment will be $\omega L^2/8$)
(Answer 1.39 MPa)



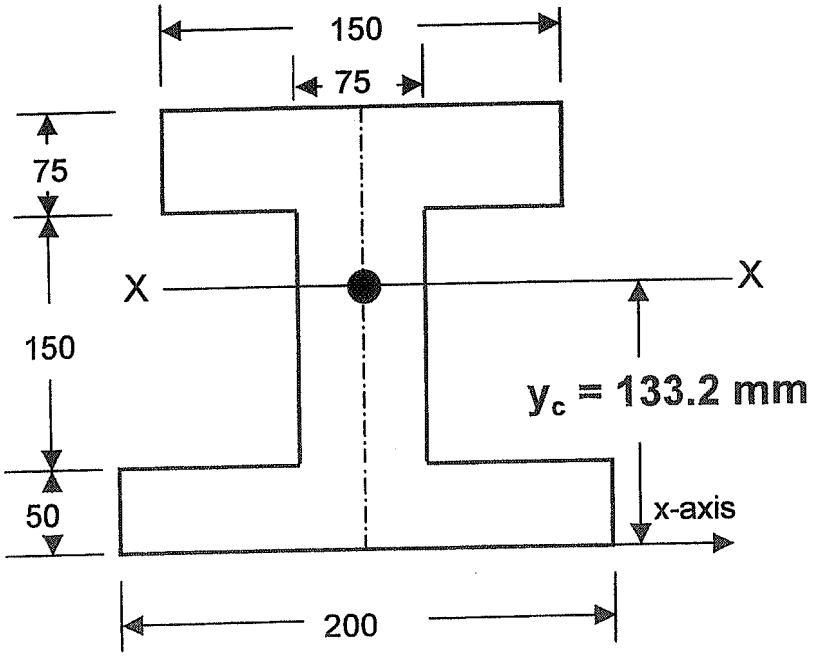
(S2.5)
A tee-bar has a flange of 1.8m, total depth of 1.2m, and uniform thickness of 12mm. The tee-bar is used as a simply supported beam of span 3.5m with the flange being horizontal. Calculate the maximum load that can be applied at mid-span if the maximum tensile stress is not to exceed 70N/mm^2. What is the maximum compressive stress in the beam?
(Hint-Maximum bending moment will be WL/4)



(Answer 1560 kN and 272 MPa)

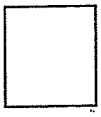
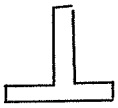
Example

A simply supported beam of I section (shown in Figure below) is simply supported at both ends with a clear span (L) of 5m. A point load (W) of 10kN acts at mid-span. Calculate the maximum stress produced in beam material due to bending.
(Hint-Maximum bending moment will be WL/4)



Solution for Tutorial 2: Second Moment of Area

(S2.1)

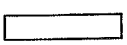
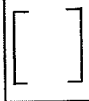
profile:	y_i	A_i	$y_i \cdot A_i$	$a_i = \lvert y_c - y_i \rvert$	$a_i^2 A_i$	I_0
	18	108	1944	7.19	5583	1296
	18	−76.96	−1385	7.19	−3979	−694
	3.28	29.6	97	7.53	1678	366
$\sum$		60.64	656		3282	968
					$I_{xx} = 4250\ \text{cm}^4$	

$y_c = 10.81\ \text{cm}$

$$z_t = \frac{I_{xx}}{e_t} = \frac{4250}{13.19}\,cm^3 = 322.2\ cm^3, \qquad e_t = 24\ \text{cm} - 10.81\ \text{cm} = 13.19\ \text{cm},$$

$$z_b = \frac{I_{xx}}{e_b} = \frac{4250}{10.81}\,cm^3 = 393.2\ cm^3, \qquad e_b = 10.81\ \text{cm}.$$

(S2.2)

profile:	y_i	A_i	$y_i \cdot A_i$	$a_i = \lvert y_c - y_i \rvert$	$a_i^2 A_i$	I_0
	25.3	−25.1	−635	12.1	−3675	—
	24.5	60	1470	11.3	7661	125
	12	64.4	772.8	1.2	93	933
	1	24	24	12.2	3572	—
$\sum$		123.3	1631.8		7651	1058
					$I_{xx} = 8709\ \text{cm}^4$	

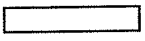
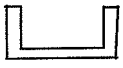
$$y_c = \frac{1631.8}{123.3}\ \text{cm} = 13.2\ \text{cm}$$

$$z_t = \frac{I_{xx}}{e_t} = \frac{8709}{13.8} cm^3 = 631\, cm^3, \qquad e_t = 27\text{ cm} - 13.2\text{ cm} = 13.8\text{ cm},$$

$$z_b = \frac{I_{xx}}{e_b} = \frac{8709}{13.2} cm^3 = 659\, cm^3, \qquad e_b = 13.2\text{ cm}.$$

(S2.3)

without reinforcements:

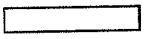
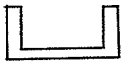
profile:	y_i	A_i	$y_i \cdot A_i$	$a_i = \lvert y_c - y_i \rvert$	$a_i^2 A_i$	I_0
	37.4	31.2	1167	20.55	13176	—
	18.9	28.64	541	2.05	122	3059
	2.36	48.3	114	14.49	10141	317
$\sum$		108.14	1822		23439	3376
					$I_{xx} = 26815\text{ cm}^4$	

$$\underline{\underline{y_c}} = \frac{1822}{108.14}\text{ cm} = \underline{\underline{16.85\text{ cm}}}$$

$$z_t = \frac{I_{xx}}{e_t} = \frac{26813}{21.15} cm^3 = 1268\, cm^3, \qquad e_t = 38\text{ cm} - 16.85\text{ cm} = 21.15\text{ cm},$$

$$z_b = \frac{I_{xx}}{e_b} = \frac{26813}{16.85} cm^3 = 1591\, cm^3, \qquad e_b = 16.85\text{ cm}.$$

with reinforcements:

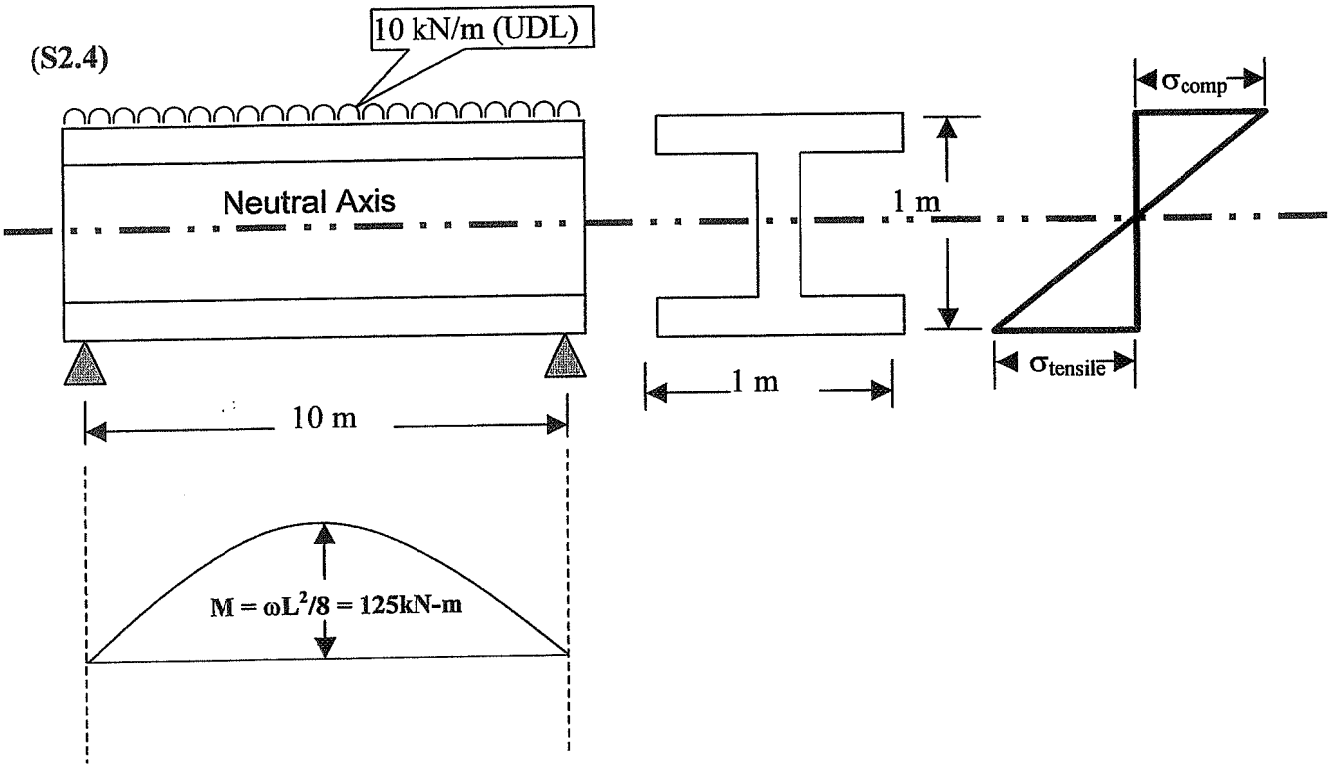
profile:	y_i	A_i	$y_i \cdot A_i$	$a_i = \lvert y_c - y_i \rvert$	$a_i^2 A_i$	I_0
	39.4	16	630	21.3	7259	—
	38.4	31.2	1198	20.3	12857	—
	19.9	28.64	570	1.8	92.8	3059
	3.36	48.3	162	14.74	10494	317
	0.5	18	9	17.6	5576	—
$\sum$		142.14	2569		36279	3376
					$I_{xx} = 39655\ \text{cm}^4$	

$$y_c = \frac{2569}{142}\ \text{cm} = 18.1\ \text{cm}$$

$$z_t = \frac{I_{xx}}{e_t} = \frac{39655}{21.7} cm^3 = 1827\ cm^3, \qquad e_t = 39.8\ \text{cm} - 18.1\ \text{cm} = 21.7\ \text{cm},$$

$$z_b = \frac{I_{xx}}{e_b} = \frac{39655}{18.1} cm^3 = 2191\ cm^3, \qquad e_b = 18.1\ \text{cm}.$$

(S2.4)



(All dimensions in Meters)

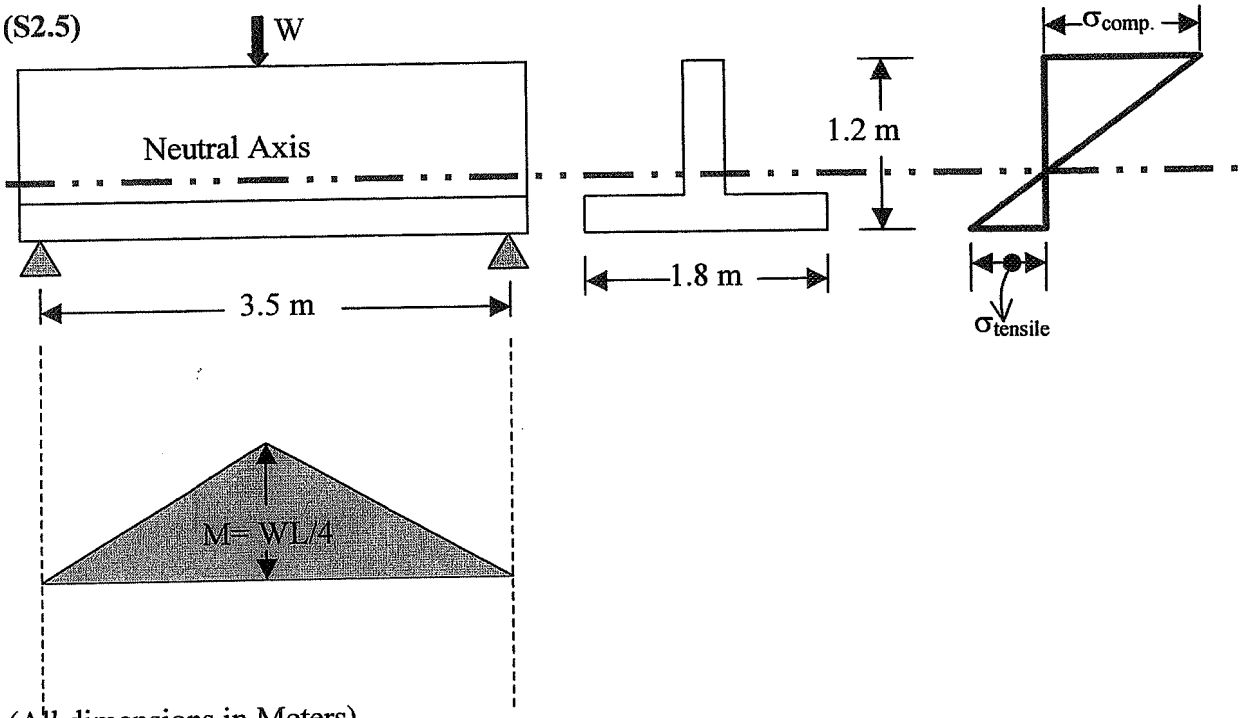
profile:	y_i	A_i	$y_i \cdot A_i$	$a_i = \lvert y_c - y_i \rvert$	$a_i^2 A_i$	I_0
	0.05	0.1	0.005	0.45	0.02025	8.33×10^{-5}
	0.5	0.08	0.04	0	0	4.26×10^{-3}
	0.95	0.1	0.095	0.45	0.02025	8.33×10^{-5}
$\sum$			0.14		0.0405	4.43×10^{-3}
					$I_{xx} = 0.045\ m^4$	

$$\underline{\underline{y_c}} = \frac{0.14}{0.28}\ m = \underline{\underline{0.5\ m}}$$

$M = \omega L^2/8 = 125\text{kN-m}$

$\sigma_{max} = M \times y_{max} / I_{xx} = 1.388\text{MPa}$

(S2.5)



(All dimensions in Meters)

profile:	y_i	A_i	$y_i \cdot A_i$	$a_i = \lvert y_c - y_i \rvert$	$a_i^2 A_i$	I_0
▭	0.006	0.0216	1.296×10^{-4}	0.238	1.22×10^{-3}	2.592×10^{-7}
▯	0.606	0.0142	8.639×10^{-3}	0.362	1.868×10^{-3}	1.676×10^{-3}
Σ			8.768×10^{-3}		3.091×10^{-3}	1.676×10^{-3}
					$I_{xx} = 0.004768\ m^4$	

$$\underline{\underline{y_c}} = \frac{0.008768}{0.03586}\ m = \underline{\underline{0.244\ m}}$$

$M = \sigma_{tensile} \times I_{xx} / y_{lower} = 70 \times 10^6 \times 0.004768 / 0.244 = 1.36 \times 10^6$ N-m

$W = 4 \times M / L = 1560$kN

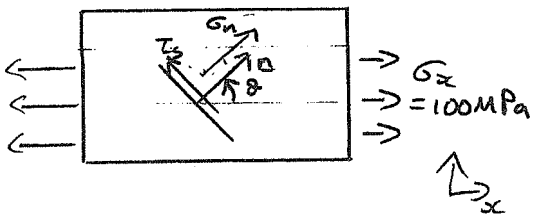
$\sigma_{compressive} = M \times y_{upper} / I_{xx} = 1.36 \times 10^6 \times 0.956 / 0.004768 = 272$ MPa

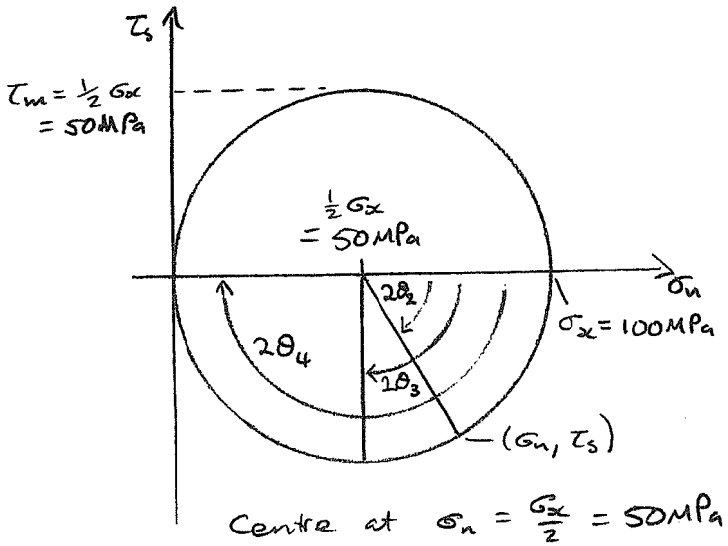
Engineering Science 23.2EI1 Strength of Materials

Solutions for tutorial sheet 3: Components of stress

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3.1/





$\theta_1 = 0°$: $\sigma_n = \sigma_x = 100MPa$

$\tau_s = 0$

Centre at $\sigma_n = \frac{\sigma_x}{2} = 50MPa$

radius $\tau_m = \frac{\sigma_x}{2} = 50MPa$

$\theta_2 = 30° \Rightarrow 2\theta_2 = 60°$:

$\sigma_n = \frac{\sigma_x}{2} + \frac{\sigma_x}{2}\cos(-60°) = 75MPa$

$\tau_s = \frac{\sigma_x}{2}\sin(-60°) = -43.3MPa$

$\Rightarrow \sigma_n = \frac{\sigma_x}{2} + \frac{\sigma_x}{2}\cos(-2\theta)$

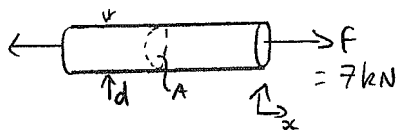
$\tau_s = \frac{\sigma_x}{2}\sin(-2\theta)$

$\theta_3 = 45° \Rightarrow 2\theta_3 = 90°$: $\sigma_n = \frac{1}{2}\sigma_x = 50MPa$, $\tau_s = \frac{1}{2}\sigma_x = 50MPa$

(by inspection of Mohr's circle)

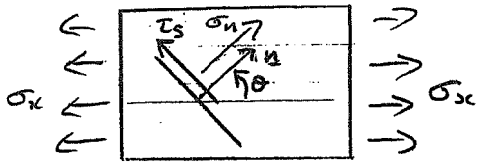
$\theta_4 = 90° \Rightarrow 2\theta_4 = 180°$: $\sigma_n = 0$, $\tau_s = 0$ (by inspection of Mohr's Circle)

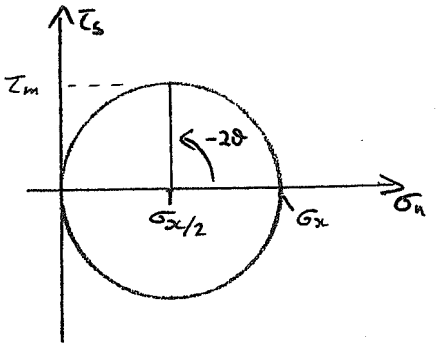
3.2/



C.S. Area $A = \pi d^2/4 = 19.6 \times 10^{-6}\ m^2$

Direct stress at yield $\sigma_{xc} = F/A = 356.5\ MPa$



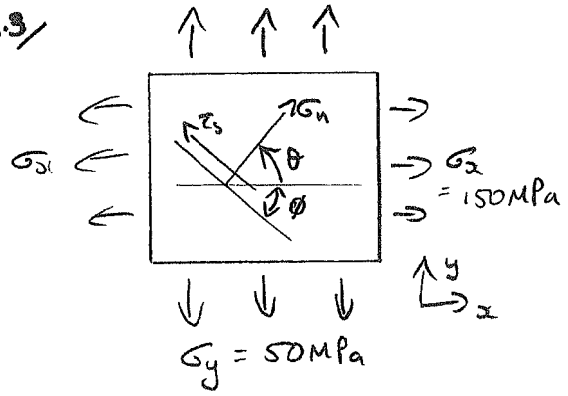


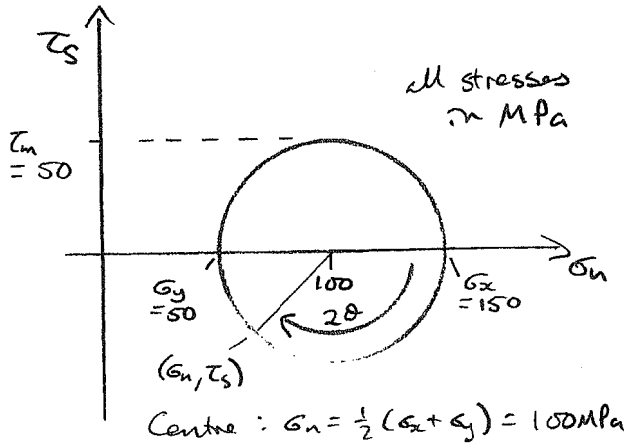
Shear stress on plane whose normal makes angle θ with x-axis is

$|\tau_s| = \frac{\sigma_{xc}}{2}|\sin 2\theta|$, which takes max value $\tau_s = \frac{\sigma_x}{2} = 178\ MPa$

for any plane with $\theta = 45°$.

3.3/





Centre : $\sigma_n = \frac{1}{2}(\sigma_x + \sigma_y) = 100 MPa$

Radius $\tau_m = \sigma_{xc} - 100 MPa = 50 MPa$

If plane is inclined at angle ϕ to x-axis,

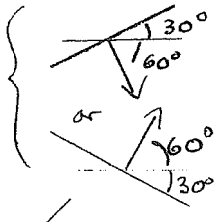
its normal makes angle $\theta = 90° - \phi$ or $\phi - 90°$ with x-axis

Sign of θ only affects sign of τ_s, so ignore.

$\phi = 30° \Rightarrow \theta = \pm 60°$: $\sigma_n = 100 + 50\cos 120° = 75\ MPa$

$2\theta = \pm 120°$

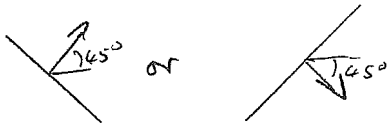
$|\tau_s| = 50 \sin 120° = 43.3\ MPa$



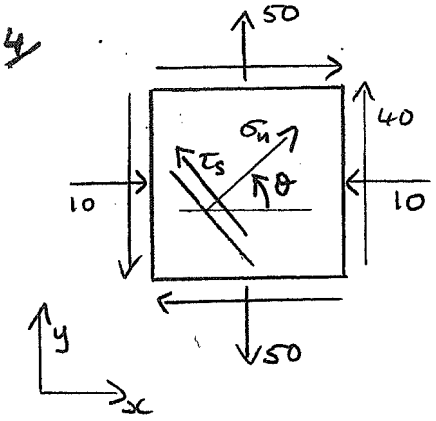
$\phi = 45° \Rightarrow \theta = \pm 45°$ $\sigma_n = 100\ MPa$

$2\theta = \pm 90°$

$|\tau_s| = 50\ MPa$



3.4



All stresses in MPa

$\sigma_x = -10$, $\sigma_y = +50$, $\tau_{xy} = +40$

$\Rightarrow$ Mohr's circle passes thro'

$(\sigma_x, \tau_{xy}) = (-10, 40)$

and $(\sigma_y, -\tau_{xy}) = (50, -40)$

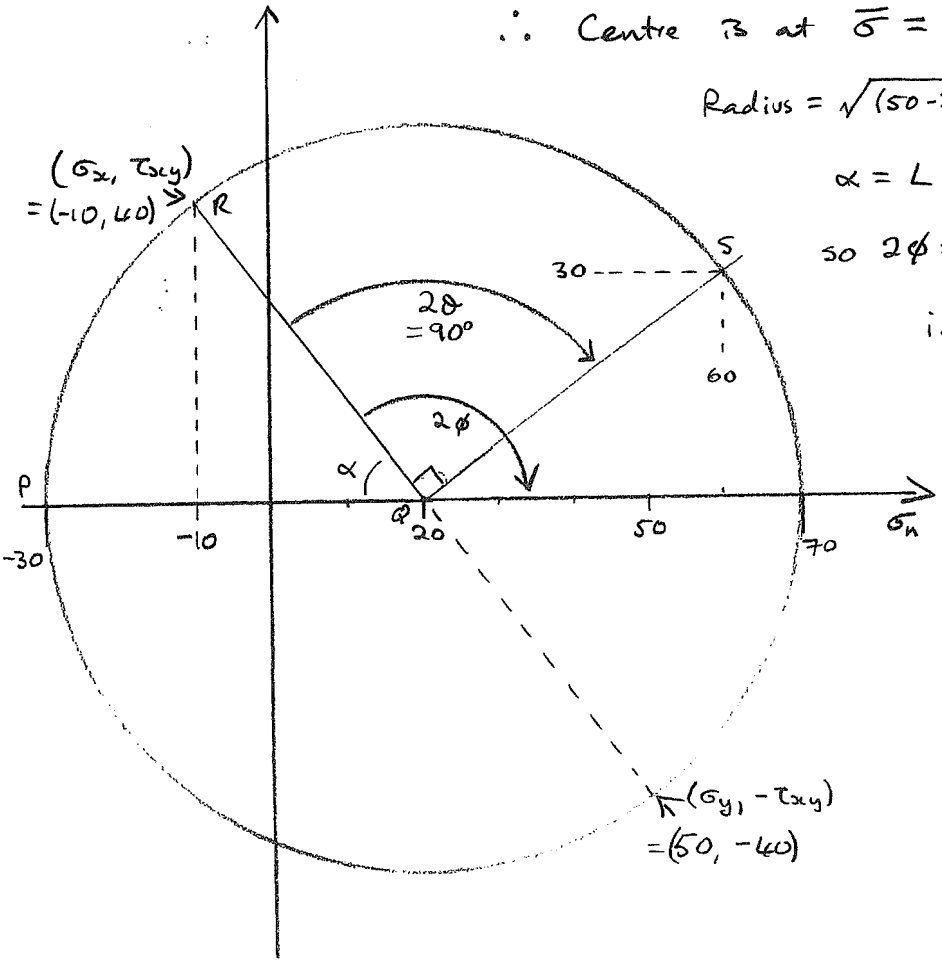
$\therefore$ Centre is at $\bar{\sigma} = \frac{1}{2}(\sigma_x + \sigma_y) = 20$

Radius $= \sqrt{(50-20)^2 + 40^2} = 50$

$\alpha = \angle PQR = \tan^{-1}\frac{40}{30} = 53.1°$

so $2\phi = 180° - \alpha = 126.9°$

i.e. $\phi = 63.4°$



a/ Shear stress is zero on planes with:

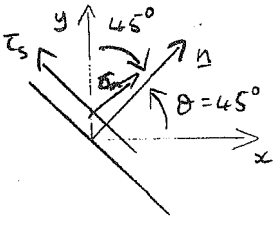
1/ $\theta = \phi = 63.4°$, where $\sigma_n = \sigma_1 = 70$ MPa

($\theta \pm 180°$, i.e. 243.4° and −116.6° are equivalent)

2/ $\theta = \phi - 90° = -26.6°$, where $\sigma_n = \sigma_2 = -30$ MPa (compressive)

($\theta = \phi + 90° = 153.4°$ is equivalent)

b/



Appropriate plane given by $\theta = 45°$, i.e. $2\theta = 90°$, shown on Mohr's circle diagram as point S:

From diagram, co-ordinates are:

$\sigma_n = 60$ MPa, $\tau_s = 30$ MPa

Solution for Tutorial 4: Second Moment of Area

(S4.1)

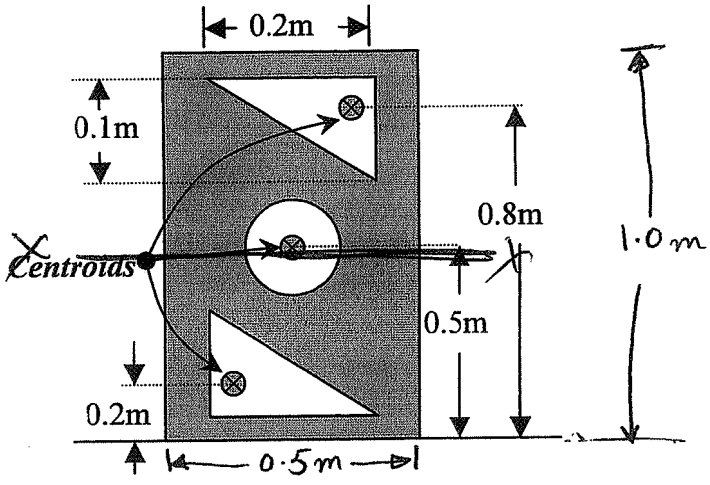


Figure 1

All dimensions in meters

profile:	y_i (m)	A_i (m^2)	$y_i \cdot A_i$ (m^3)	$a_i = \lvert y_c - y_i \rvert$ (m)	$a_i^2 A_i$ (m^4)	I_0 (m^4)
□	0.5	$0.5\times1 = +0.5$	$+0.25$	0	0	$\frac{bh^3}{12} = \frac{0.5\times1^3}{12} = +0.04166$
◺	0.2	$-\frac{0.1\times0.2}{2} = -0.01$	-2×10^{-3}	$\lvert0.5-0.2\rvert = 0.3$	-9×10^{-4}	$-\frac{bh^3}{36} = -\frac{0.2\times(0.1)^3}{36} = -5.55\times10^{-6}$
○	0.5	$-\pi\times(0.1)^2 = -0.0314$	-0.0157	0	0	$-\frac{\pi R^4}{2} = -\frac{\pi\times(0.1)^4}{2} = -1.57\times10^{-4}$
◺	0.8	$-\frac{0.1\times0.2}{2} = -0.01$	-8×10^{-3}	$\lvert0.5-0.8\rvert = 0.3$	-9×10^{-4}	$-\frac{bh^3}{36} = -5.55\times10^{-6}$
Σ		0.4486	0.2243		-1.8×10^{-3}	0.04149

$$y_c = \frac{1}{\Sigma A_i} \times \Sigma y_i A_i = \frac{0.2243}{0.4486}$$

$$I_{xx} = \Sigma(a_i^2 A_i + I_0) = 0.0397\ m^4$$

$$\underline{\underline{y_c = 0.5\ m}}$$

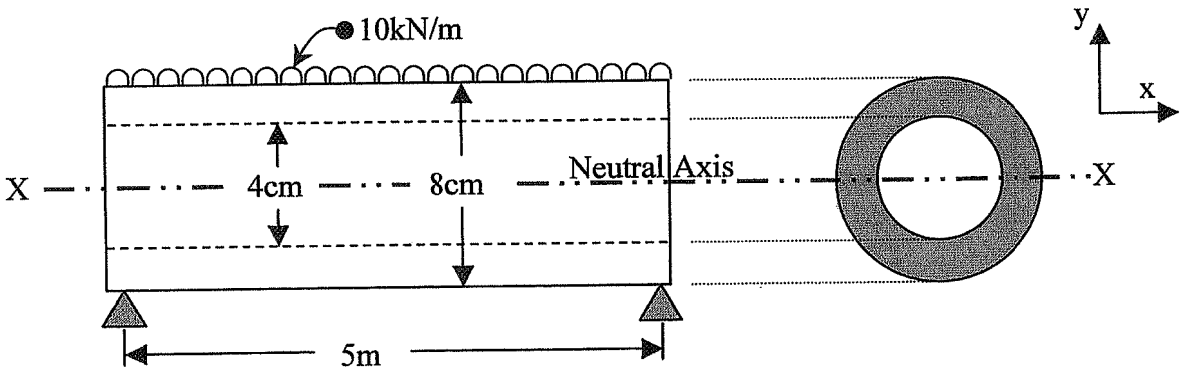
$$z_t = \frac{I_{xx}}{e_t} = \frac{0.039\ 7}{0.5} m^3 = 0.0794\ m^3, \qquad e_t = 0.5\ m,$$

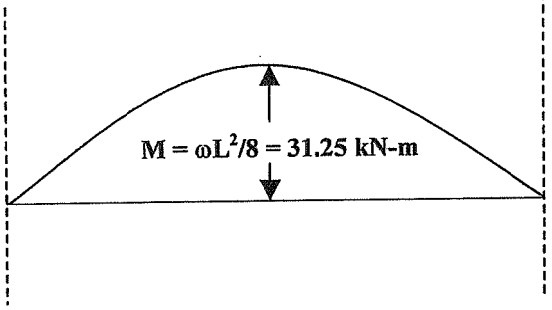
$$z_b = \frac{I_{xx}}{e_t} = \frac{0.039\ 7}{0.5} m^3 = 0.0794\ m^3, \qquad e_b = 0.5\ m.$$

Note that in this solution the centroids of △'s + ○ are not on any axis of symmetry but we can still apply // axis theorem as the distance w.r.t x-axis or xx is fixed.

(S4.2)

a)





$$M_{max} = \frac{\omega L^2}{8} = \frac{10 \times 25}{8}$$

$$= 31.25 \text{ kN-m}$$

(Note that centroidal axis is xx and not zz)

$$I_{xx} = \frac{\pi R^4}{4} \quad \text{for circle}$$

$R_1 = 8\text{cm}$

$R_2 = 4\text{cm}$

$$\Rightarrow I_{xx} \text{ for Hallow Shaft(Beam)} = \frac{\pi}{4}\left\{R_1^4 - R_2^4\right\} = \frac{\pi}{4}\left\{(0.04)^4 - (0.02)^4\right\}$$

$$= 1.885 \times 10^{-6} \ m^4$$

$$\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$$

$\because y_{max} = R_1$

$$\Rightarrow \sigma_{max} = \frac{M_{max} \times y_{max}}{I_{xx \text{ for circular}}} = \frac{31.25 \times 10^3 \times 0.04}{1.885 \times 10^{-6}} = 663 \text{ MPa}$$

$$Z_{1,2} = \frac{I}{y_{1,2}} = \frac{1.885 \times 10^{-6}}{0.04} = 4.7125 \times 10^{-5} \ m^3$$

(S4.2) Solution continued

b) If replaced by solid shaft of same area of cross-section.
Lets assume that "R" is the radius of the new solid shaft. Hence for above condition to be satisfied:

$$\pi R^2 = \pi (R_1^2 - R_2^2)$$

$$\Rightarrow R = \sqrt{(0.04)^2 - (0.02)^2} = 0.03464$$

$$\Rightarrow I_{xx} \text{ for Solid Shaft (Beam)} = \frac{\pi}{4} R^4 = \frac{\pi}{4} (0.03464)^4$$

$$= 1.13 \times 10^{-6} \ m^4$$

$y_{max} = R$

$$\Rightarrow \sigma_{max} = \frac{M_{max} \times y_{max}}{I_{xx \text{ for Solid}}}$$

$$= \frac{31.25 \times 10^3 \times 0.03464}{1.13 \times 10^{-6}} = 958 \ MPa$$

c) Section modulus of Solid beam $= \frac{1.13 \times 10^{-6}}{0.03464} = 3.26 \times 10^{-5} \ m^3$

is lower than that of Hollow beam and hence higher Stress in Solid beam of same cross-sectional area. Hence best to shift mass of shaft away from centroid e.g. in this case by making it hollow. Typical applicati eg in bicycle forks! to get max strength to weight ratio.

(S4.3)

Note: This is summary solution and students are advised to follow the procedure of Mohr's Stress circle illustrated in the class.

Brief solution: You should show more working in exam

all stresses in MPa

$\alpha = 30^\circ \Rightarrow \theta = 60^\circ \Rightarrow 2\theta = 120^\circ \Rightarrow \sigma_n = 50 - 150 \cos 360 = 0$

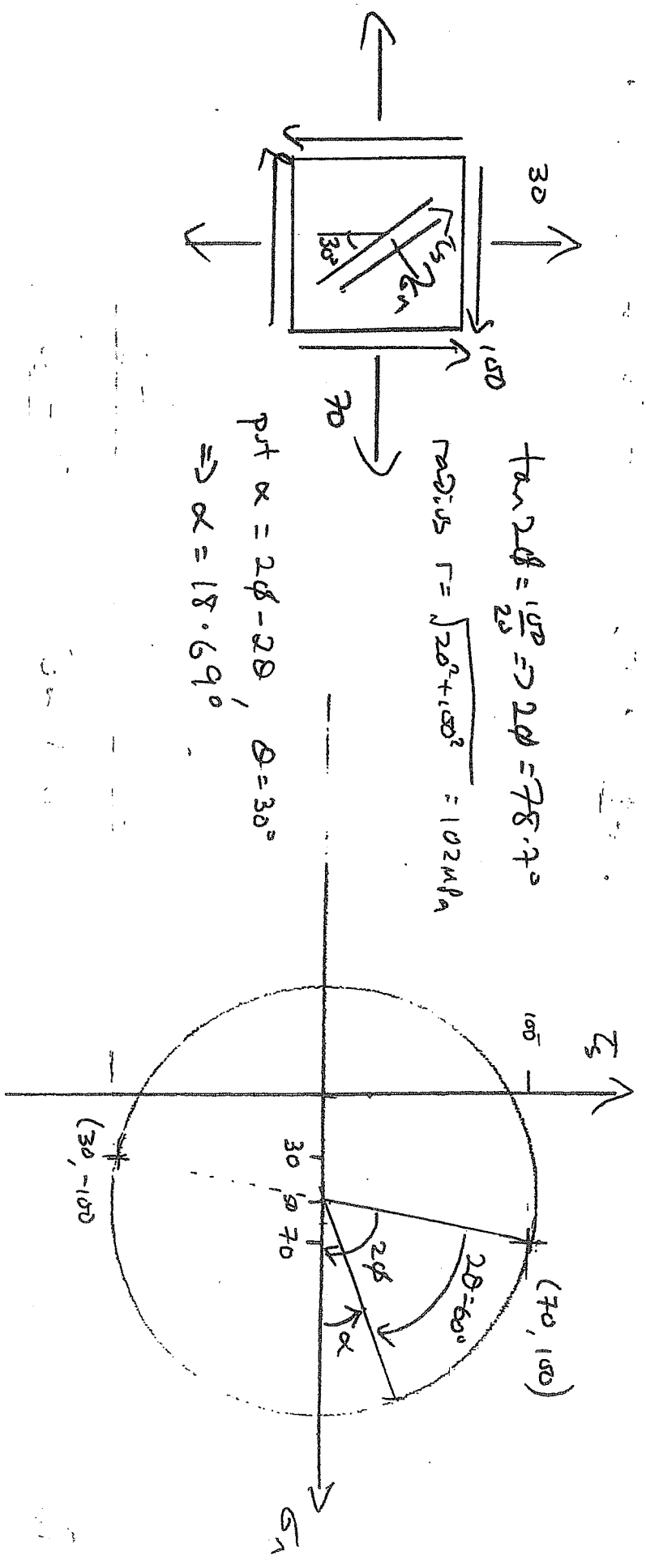
$\tau_s = -100 \sin 60^\circ = -86.7\ MPa$

(or $+86.7\ MPa$ if you take β)

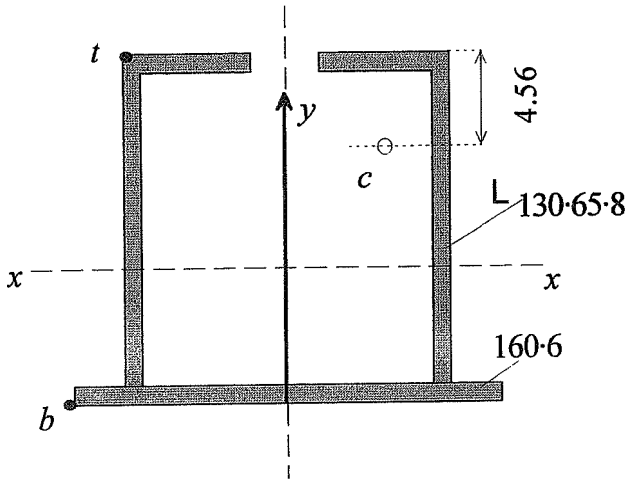
$\alpha = 45^\circ \Rightarrow \theta = 45^\circ \Rightarrow 2\theta = 90^\circ \Rightarrow \sigma_n = 50\ MPa, \quad \tau_s = -100\ MPa$ (or $+$)

(S4.4)

Note: This is summary solution and students are advised to follow the procedure of Mohr's Stress circle illustrated in the class.



(S4.5)



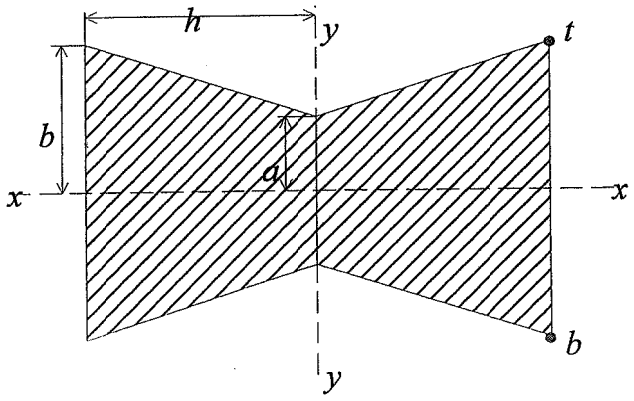
profile:	y_i	A_i	$y_i \cdot A_i$	$a_i = \lvert y_c - y_i \rvert$	$a_i^2 A_i$	I_0
▭	0.3	9.6	3	6.61	419	—
⌜ ⌝	9.04	29.9	270	2.13	136	526
$\sum$		39.5	273			
					$I_{xx} = 1081\ \text{cm}^4$	

$$\underline{\underline{y_c}} = \frac{273}{39.5}\ \text{cm} = \underline{\underline{6.93\ \text{cm}}},$$

$$\underline{\underline{Z_t}} = \frac{I_{xx}}{e_t} = \frac{1081}{6.67}\ cm^3 = \underline{\underline{162\ cm^3}}, \qquad e_t = 13.6\ \text{cm} - 6.93\ \text{cm} = 6.67\ \text{cm},$$

$$\underline{\underline{Z_b}} = \frac{I_{xx}}{e_b} = \frac{1081}{6.93}\ cm^3 = \underline{\underline{156\ cm^3}}, \qquad e_b = 6.93\ \text{cm}.$$

(S4.6)



By virtue of steiner's Theorem we may write:

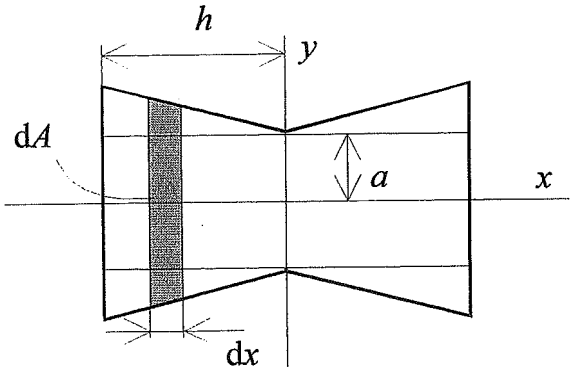
$$I_{xx} = I_{\mathrm{Sq}} + 4\left(I_{\mathrm{Tr}} + s_y^2 A_{\mathrm{Tr}}\right), \qquad s_y = a + \frac{1}{3}(b-a),$$

$$I_{\mathrm{Sq}} = \frac{2h(2a)^3}{12},$$

$$I_{\mathrm{Tr}} = \frac{h(b-a)^3}{36}, \qquad A_{\mathrm{Tr}} = \frac{h(b-a)}{2}.$$

Consequently:

$$\underline{\underline{I_{xx}}} = \frac{4}{3}ha^3 + 4\left(\frac{h(b-a)^3}{36} + \left(\frac{2}{3}a + \frac{1}{3}b\right)\frac{h(b-a)}{2}\right) = \underline{\underline{\frac{1}{3}h\left(a^3 + b^3 + ab^2 + a^2 b\right)}}.$$



$$\underline{\underline{I_{yy}}} = \int_A x^2 \mathrm{d}A = \int_{-h}^{+h} x^2 y(x)\,\mathrm{d}x = \int_{-h}^{0} x^2 2\mathrm{d}x\left(a + \frac{b-a}{-h}x\right) + \int_0^h x^2 2\mathrm{d}x\left(a + \frac{b-a}{h}x\right) = \underline{\underline{h^3\left(\frac{1}{3}a + b\right)}},$$

$$Z_{xt} = \frac{I_{xx}}{b} = \frac{1}{3}\frac{h}{b}\left(a^3 + b^3 + ab^2 + a^2 b\right),$$

$$Z_{yt} = \frac{I_{yy}}{h} = h^2\left(\frac{1}{3}a + b\right).$$

In particular, for the case $b = 2a$ it follows that:

$$I_{xx} = \frac{1}{3}h\left(a^3 + 8a^3 + 4a^3 + 2a^3\right) = 5ha^3,$$

$$I_{yy} = h^3\left(\frac{1}{3}a + 2a\right) = \frac{7}{3}h^3a,$$

and if:

$$I_{xx} = I_{yy}$$

then:

$$h = \sqrt{\frac{15}{7}}\,a.$$